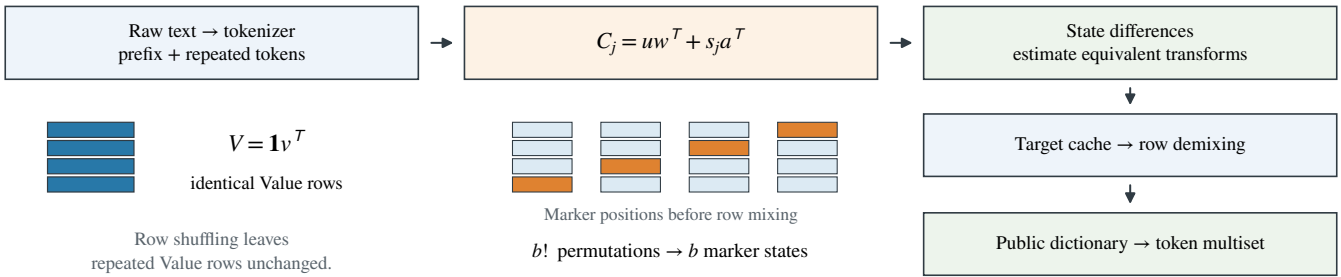
(a) Repeated-token input
Raw text → tokenizer
prefix + repeated tokens
$V = \mathbf{1}v^T$
identical Value rows
Row shuffling leaves
repeated Value rows unchanged.
(b) Observable marker orbit
$C_j = uw^T + s_j a^T$
Marker positions before row mixing
$b!$ permutations → b marker states
(c) Transfer within a key epoch
State differences
estimate equivalent transforms
Target cache → row demixing
Public dictionary → token multiset